Cropping Cycles from Space

Sentinel-2 NDVI and CHIRPS rainfall for the cropland of Bojonegoro, East Java, 2019–2024 | Google Earth Engine + R | Self-initiated portfolio project using public data

The question

How many crops per year are grown on Bojonegoro's cropland, when are they planted, and how closely does the cropping calendar follow the rains? Field calendars are rarely available for a whole regency, but six years of Sentinel-2 imagery can reconstruct them from the greening and browning of the land.

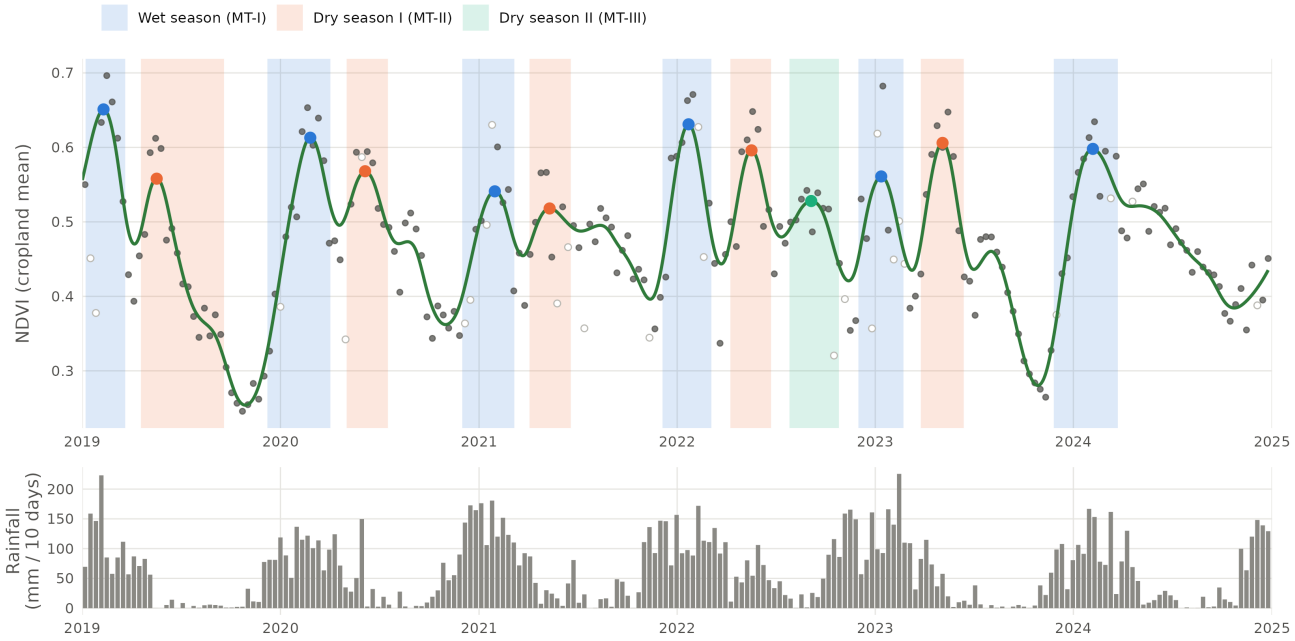
Data and methods

Step	What was done
Imagery	Sentinel-2 surface reflectance 2019–2024; clouds masked with Cloud Score+ (threshold 0.60)
Indices	NDVI (greenness) and NDWI (NIR–SWIR, wetness); 10-day median composites
Area	Bojonegoro Regency (2,313 km²); cropland only (ESA WorldCover 2021, 949 km²)
Rainfall	CHIRPS daily rainfall summed per 10-day period
Quality control	Periods with < 50% cloud-free cropland excluded (26 of 219, mostly Dec–Feb)
Smoothing	Weighted smoothing spline (weights = cloud-free share), RMSE 0.033 NDVI
Crop cycles	NDVI peaks ≥ 0.03 above surrounding troughs, ≥ 60 days apart; start/end at 20% of amplitude (TIMESAT-style threshold)
Rain onset	First 10-day period from 1 Sept with ≥ 50 mm, followed by ≥ 50 mm in the next 20 days

Result 1: two main crop cycles per year, timed by the rains

Bojonegoro's cropland greens up twice a year, following the rains

Grey dots: 10-day Sentinel-2 composites (hollow = cloudy, excluded). Green line: weighted smoothing spline. Shading: detected crop cycles.



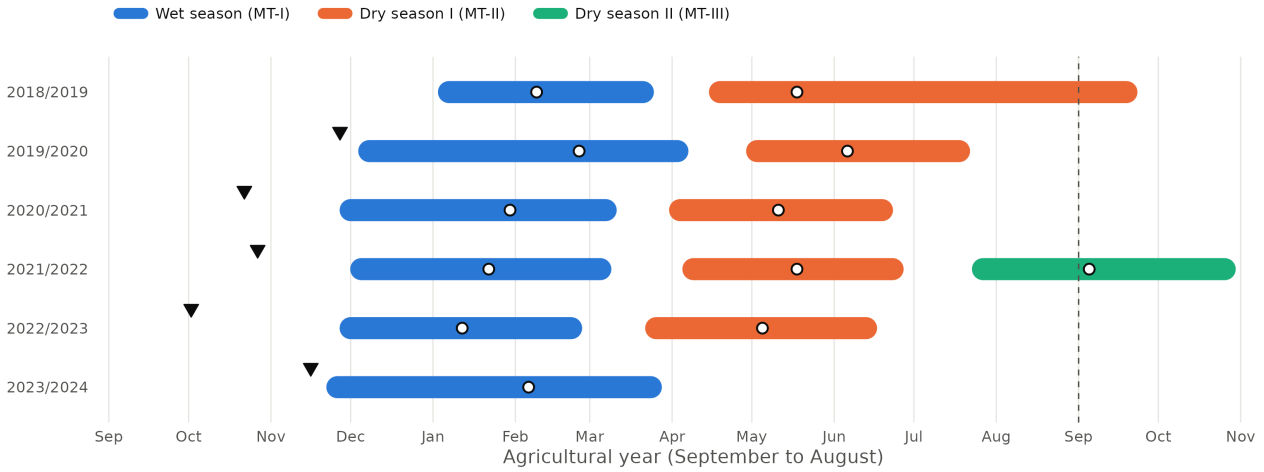
Data: Copernicus Sentinel-2 (Cloud Score+ masked), CHIRPS rainfall, ESA WorldCover 2021 cropland, geoBoundaries. Self-initiated portfolio project using public data.

- The **wet-season crop (MT-I)** peaks around late January (median peak NDVI 0.61); the **first dry-season crop (MT-II)** peaks around May (median 0.57).
- A **third crop (MT-III)** appeared only in 2022, the wettest year of the record (about 2,780 mm).

Result 2: a stable planting window

The wet-season crop starts between late November and early December every year

Bars: start to end of each detected crop cycle (20% of NDVI amplitude). White dot: NDVI peak. Black triangle: rainy-season onset.



2018/2019 begins in January 2019 (start of the record). Dashed line: end of the agricultural year.
Rainy-season onset: first 10-day period from 1 September with >= 50 mm, followed by >= 50 mm in the next 20 days.

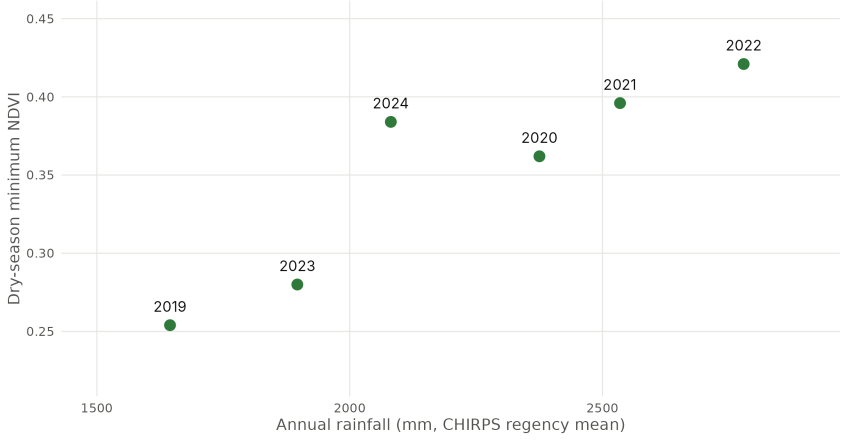
Season	Rainy-season onset	Wet-season green-up	Days after onset
2019/2020	27 Nov 2019	8 Dec 2019	11
2020/2021	22 Oct 2020	1 Dec 2020	40
2021/2022	27 Oct 2021	5 Dec 2021	39
2022/2023	2 Oct 2022	1 Dec 2022	60
2023/2024	16 Nov 2023	26 Nov 2023	10

The wet-season crop started between 26 November and 8 December every year, although the rains began anywhere between early October and late November. When the rains came early, farmers waited; in the two late-onset years (2019 and 2023) green-up followed within about 10 days. Planting appears to be timed by an established local calendar and by sufficient soil moisture, rather than by the first rains alone.

Result 3: dry years, browner dry seasons

In drier years, cropland turned browner in the dry season

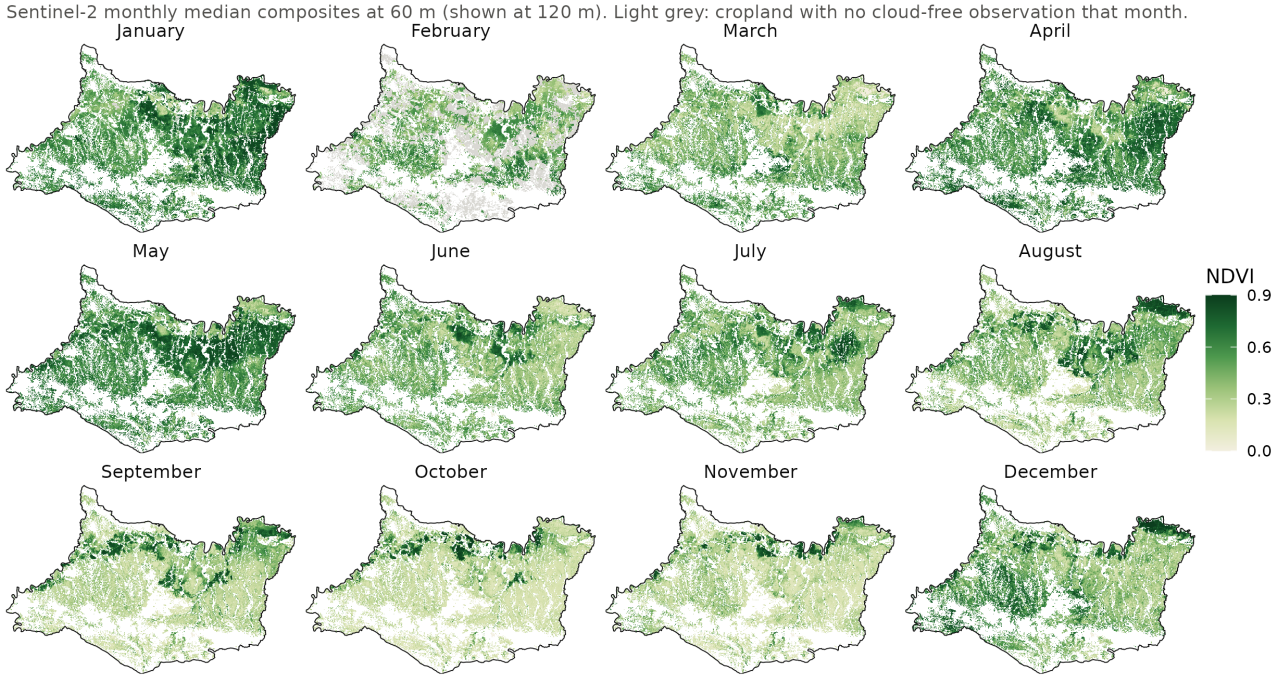
Lowest NDVI from August to December vs annual rainfall (6 years; Spearman rho = 0.94)



2019 and 2023 were dry years (2019: strong positive Indian Ocean Dipole; 2023: El Nino).

Dry-season minimum NDVI was 0.25 in 2019 and 0.28 in 2023, against 0.36–0.42 in the other years (Spearman ρ = 0.94 with annual rainfall, n = 6). 2019 had a strong positive Indian Ocean Dipole and 2023 an El Niño. With six years this is descriptive, not a trend.

Result 4: the 2023 cropping year in maps



Crops are green in January and April-May and brown from September to November 2023; fields in the northern river valley stay green longer.

Cropland was green in January and April–May, then browned from September to November. Fields in the northern river valley stayed green longer, which suggests access to water during the dry season. February had the most cloud: 40% of cropland had no cloud-free observation that month.

Limitations

- The series is a regency-wide average over mixed crops (rice, maize, tobacco and others); overlapping planting dates can blur cycles, as in the long dry-season plateaus of 2021 and 2024. Pixel-level phenology or a crop-type map would separate them.
- NDVI is an indicator of crop growth, not of yield or planted area, and was not validated with field data.
- Cloud removes 12% of periods, mainly at the height of the wet season.

What a client receives in this kind of project

- A commented Earth Engine script that can be re-run for any area and period.
- Clean time series (CSV), crop-cycle and rainfall tables, and a commented R script.
- Time-series charts, a crop calendar and monthly maps, plus a short interpretation note with limitations.

Data: Copernicus Sentinel-2 (modified Copernicus Sentinel data 2019–2024); Cloud Score+ (Pasquarella et al. 2023); CHIRPS (Funk et al. 2015); ESA WorldCover 2021 (CC BY 4.0); geoBoundaries v6 (CC BY 4.0). Software: Google Earth Engine; R 4.3.3 with ggplot2, patchwork, terra and sf.